

Integration by Substitution

Date _____ Period _____

Evaluate each indefinite integral. Use the provided substitution.

1) $\int \frac{20x^3}{\sqrt{25 - 25x^8}} dx; \quad u = 5x^4$

$$\sin^{-1} \frac{5x^4}{5} + C$$

2) $\int \frac{10x^4}{9 + 4x^{10}} dx; \quad u = 2x^5$

$$\frac{1}{3} \cdot \tan^{-1} \frac{2x^5}{3} + C$$

3) $\int -\frac{2 \cdot \csc^2 2x}{\cot(2x) \cdot \sqrt{\cot^2 2x - 1}} dx; \quad u = \cot 2x$

$$\sec^{-1} |\cot 2x| + C$$

4) $\int \frac{1}{x\sqrt{25 - (\ln -2x)^2}} dx; \quad u = \ln -2x$

$$\sin^{-1} \frac{\ln -2x}{5} + C$$

Evaluate each indefinite integral.

5) $\int \frac{8x}{\sqrt{9 - 16x^4}} dx$

$$\sin^{-1} \frac{4x^2}{3} + C$$

6) $\int \frac{3x^2}{x^3 \sqrt{x^6 - 1}} dx$

$$\sec^{-1} |x^3| + C$$

7) $\int \frac{10x}{16 + 25x^4} dx$

$$\frac{1}{4} \cdot \tan^{-1} \frac{5x^2}{4} + C$$

8) $\int -\frac{4\sin 4x}{\sqrt{9 - \cos^2 4x}} dx$

$$\sin^{-1} \frac{\cos 4x}{3} + C$$